



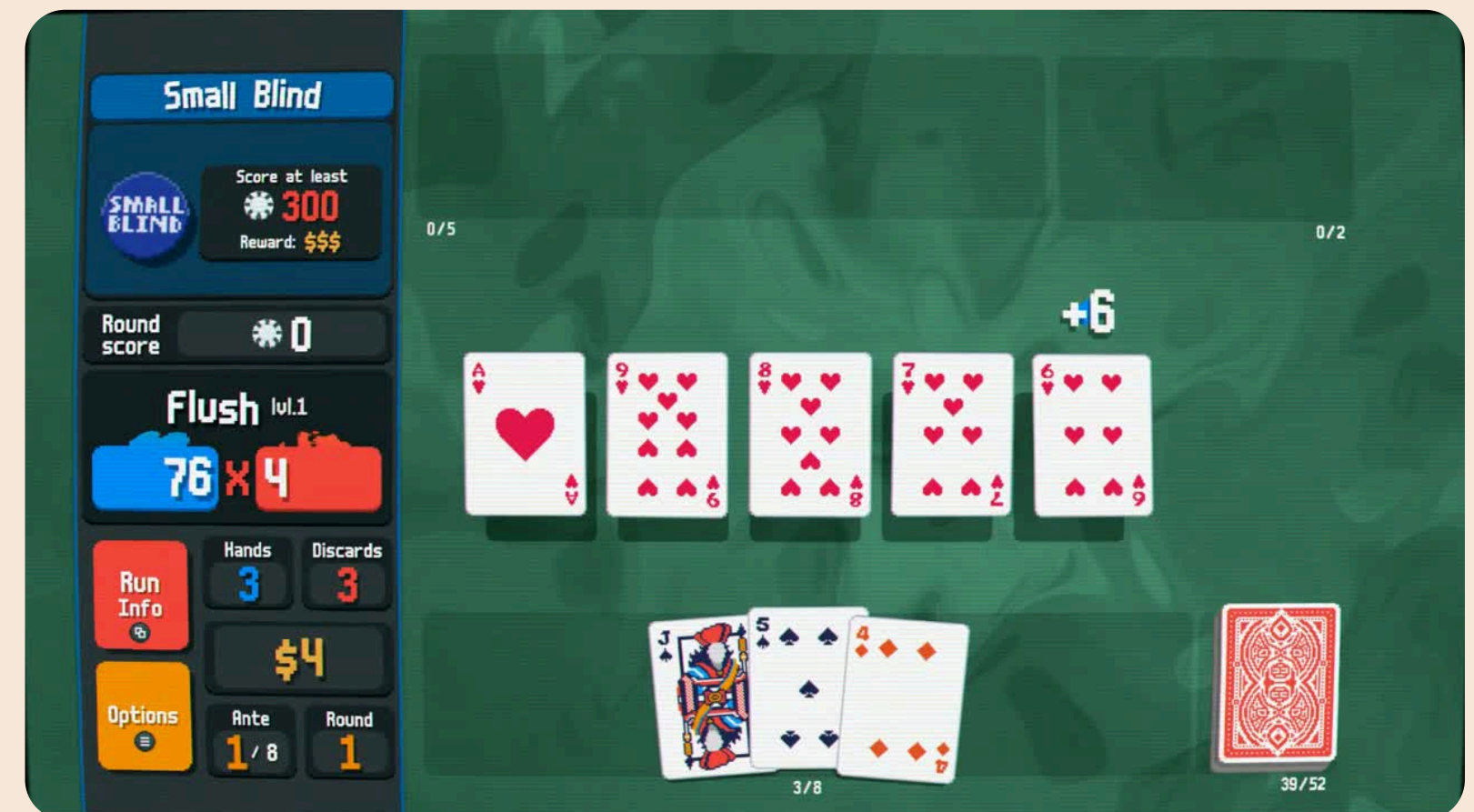
RL AND STRATEGY BASED MODEL TO PLAY BALATRO

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WHAT IS BALATRO?

- **Core Concept:** A roguelike deckbuilding card game where players must score enough points to pass increasingly difficult rounds called "blinds".
- **Gameplay Mechanic:** Points are earned by playing standard poker hands like pairs, flushes, or full houses.
- **The "Joker" System:** The game features 150 unique Jokers that modify scoring (e.g., awarding +30 chips for a flush or doubling the total score).
- **Strategy:** Players hold up to 5 Jokers at a time and must acquire them through a shop between rounds.



THE CORE PROBLEM

- **Context-Dependence:** A Joker's value isn't fixed; it depends entirely on what other jokers you have.
- **Long-Horizon Planning:** Success requires a sequence of 24 rounds where early choices constrain your future.
- **The Gap:** Current RL agents are greedy and they fail to build a strategy to reach higher antes.

Ante	Blind	Target
1	Small	350
1	Big	450
1	Boss	600
2	Small	800
2	Big	1200
2	Boss	1600
4	Boss	12000
5	Small	20000

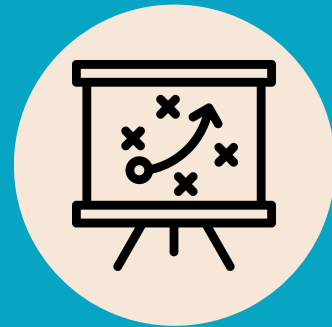
Core question: Would an agent that just learns to evaluate Joker-Strategy synergy, delegating card selection to an algorithm, perform better?

ENVIRONMENT CLASSIFICATION

Property	Classification	Why
Observability	Partially observable	Future shop offers / Joker draws hidden
Determinism	Deterministic (seeded)	Fixed seed
Agency	Single-agent	No opponent
Dynamics	Static (turn-based)	Environment waits for the agent

Simple classification, but 150 Jokers × deck composition × shop economy = a large, non-trivial decision space.

TWO-TIER ARCHITECTURE



Strategic (long-term)

pick a high-level build
(flush / mult / pair)
buy appropriate jokers

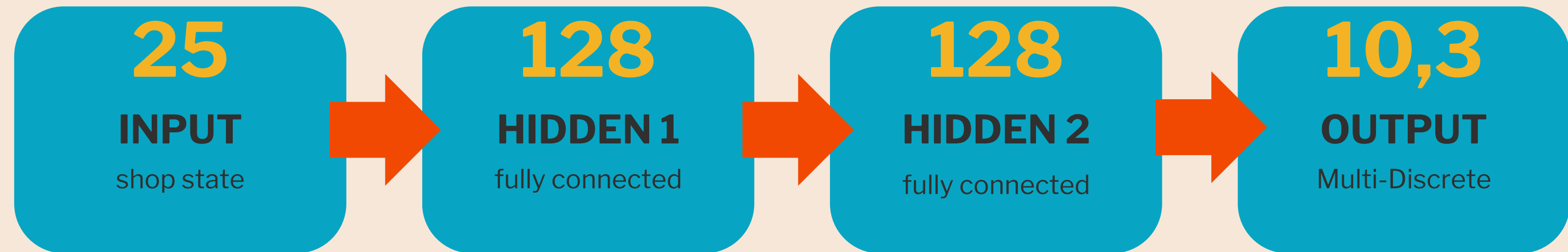


Operational (short-term)

card playing
and discarding

Inspired by how expert players actually play: commit to a strategy, choose cards and buy jokers according to it.

THE ARCHITECTURE OF THE RL MODEL



- The RL model is based on the **PPO Algorithm**.
- **Input layer:** 25 units: Economy signals (ante, round, money, blind target), Joker slot occupancy (5 slots $\times$ 2 = 10), shop card identity (normalized enum)/cost (4 slots $\times$ 2 = 8), and the active strategy as a 3-dim one-hot (so it is using 3 units, not one).
- **Two 128-unit hidden layers:** We felt like it would be big enough to represent Joker/shop interactions, small enough to train fast during rapid iteration. We increased them from 64 to 128 after adding joker selection to accommodate for anticipated complexity.
- **Output layer:** 10 and 3 units MultiDiscrete: 1 skip + 4 buy (shop slots) + 5 sell (owned Joker slots) and 3 for strategy selection.

REWARD FUNCTION

The PPO model repeatedly plays Balatro, updating its policy to favor actions that increase expected rewards for surviving to higher antes.

The rewards themselves get higher as every ante is harder to beat than the last one.

When shop decisions were introduced, we also introduced a reward for buying a joker. But then we scrapped it because we observed that it wasn't needed.

Ante Reached / Action	Reward
Game over (ante 1)	-0.5
Ante 2	+0.2
Ante 3	+0.4
Ante 4	+0.6
Ante 5	+0.8
Ante 6	+1.0
Ante 7	+1.5
Ante 8 and beyond	+2.0
Selling joker	-0.5

DECISION POINT #1

PLATFORM: balatro-gym → Balatrobot

Initial Approach:

- balatro-gym (existing Gymnasium environment) would be our foundation.

Decision:

- Abandon balatro-gym after intense technical assesment.
- Adopt Balatrobot v1.4.1 - stable RPC control of the real game engine.
- Create a simplified and controlled environment suitable for training.

MOD

- Balatro has a huge state space (150 jokers, 12 planets, 22 tarot cards, 18 spectral cards, 5 joker editions, ...)
- The mod reduces this to just 14 jokers and nothing else
- Max capacity of booster packs, vouchers, and consumables in shop set to 0, so they cannot spawn
- Jokers are stripped of any editions before appearing in the shop
- Boss blind effects disabled: some by removing their effects (disabling debuffs) and some by modifying game logic (cards in hand can never be face down)

This simplified environment made training more manageable

DECISION POINT #2

FROM STRATEGY SELECTION TO SHOP MANAGEMENT

Initial design:

- Agent picks a strategy label (flush / pair / mult) once per ante.
- A hand-written heuristic handles card play without ever discarding, conditioned on the strategy.
- 2x64 Hidden Layers

What changed:

- Agent now manages both the strategy decisions and the shop decisions: buy / sell / skip.
- Card play stays deterministic; Monte Carlo added for discard decisions.
- 2x128 Hidden Layers

MONTE CARLO DISCARD IMPLEMENTATION

Problem: deterministic calculator can pick the best hand from cards already drawn, but can't evaluate whether discarding is better.

Solution: Monte Carlo estimation - simulate N draws per candidate discard, compare expected value of discarding vs. playing.

Known limitation: This algorithm used to make training 6× slower for 50 simulations; we optimized it using a caching mechanism for the the combinations already calculated in the last play, and also by pruning the the combinations that weren't worthy of calculating.

INFRASTRUCTURE

MAKING IT FAST ENOUGH TO TRAIN

4.80x

**Naive episode
reset**

load()-based reset
instead of start():
(0.049s vs 0.233s avg)

4

**Parallel
instances**

near-linear scaling

~8HRS

**100,000
episodes**

with 4 parallel instances -
overnight-training
feasible

- The time to train dropped from ~36 hours to ~8 hours

DECISION POINT #3

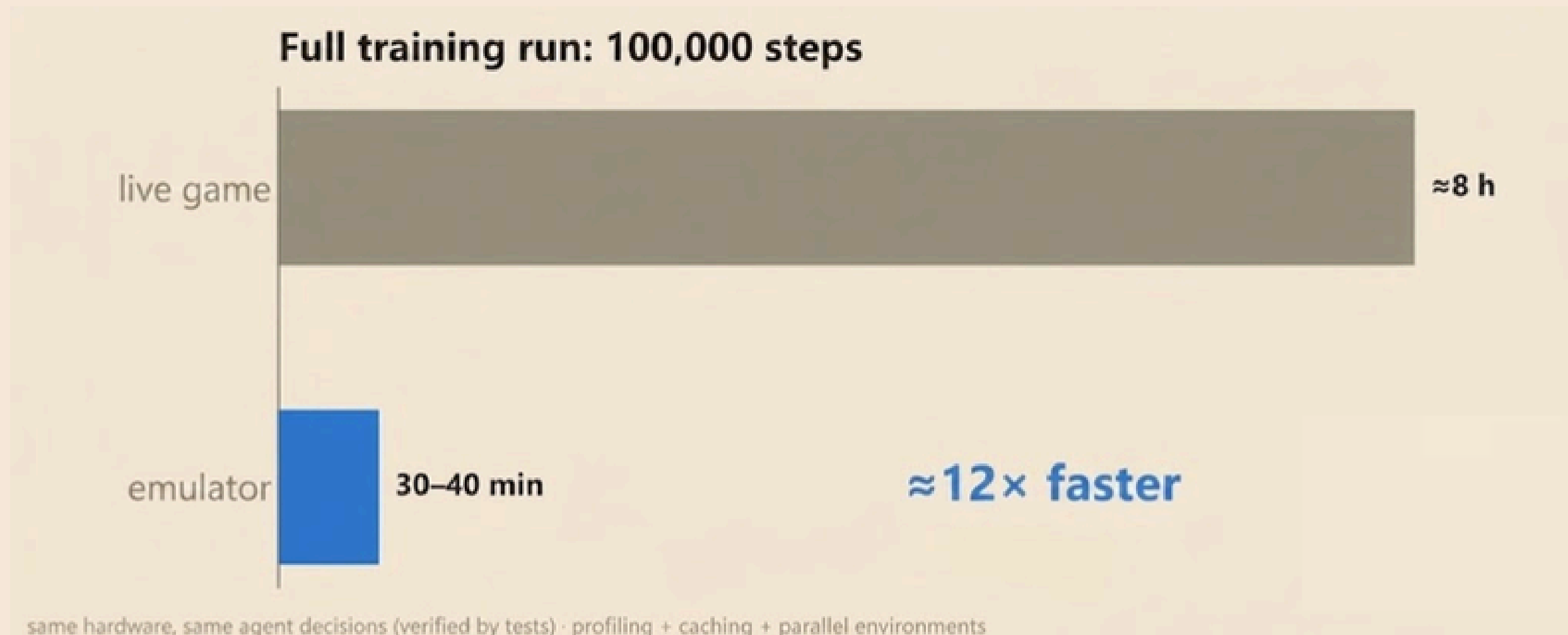
INTEGRATING AN EMULATOR

Training an RL agent means playing millions of hands. The real game is too slow.

- Balatro is a video game, it renders graphics, runs animations, and only exists to be played by humans, one instance at a time
- RL training needs: thousands of games per hour, run in parallel, no graphics, restartable in milliseconds
- Standard solution in RL research: replace the game with an emulator, a program that reproduces the game's rules and math, but nothing it doesn't need (no visuals, no delays)
- Integrated the jackdaw Balatro emulator: full game rules, runs in terminal: no game client, no network
- Reproduces the real game's mechanics: scoring ($\text{Chips} \times \text{Mult}$), blinds, shop, joker effects, deck randomness
- Speaks the same protocol as the real game's bot interface → our training code doesn't know the difference; one flag switches between real game and emulator

RESULTS AFTER INTEGRATION OF THE EMULATOR

- Increased the speed of training to manageable levels
- Seeded & deterministic → every experiment is exactly reproducible
- For the first time we could afford to experiment on the full, non-simplified game environment



WHAT WE IMPLEMENTED

REPLAY SYSTEM: SEEING THE AGENT PLAY

- Web dashboard: browse all training episodes, sorted by score, ante reached, win/loss
- Step-by-step timeline of one game: what the agent bought, what strategy it declared, which hands it played, which jokers it held at that moment

The screenshot displays a web dashboard for a poker training system. At the top right, it indicates 'Step 6 / 99 - 2 hands'. The dashboard is divided into several sections:

- ACTIVE JOKERS:** Shows 1/5 active jokers, with 'Jolly Joker' selected.
- EPISODES:** Lists 6147 episodes. A table shows episode details: Episode #, Score, Ante, and Time. The first four episodes are #1275 with scores +5.8, +6.8, +6.9, and +13.8, all with ante A4-R11 or A4-R12. The fifth episode is #1276 with a score of +13.8 and ante A6-R17.
- HAND HISTORY:** Shows two hands played. Hand 1 (A1, R1, H2) is a 'flush' with cards A♠, Q♠, T♠, 7♠, 4♠. Hand 2 (A1, R1, H1) is a 'flush' with cards 9♠, 7♠, 6♠, 5♠, 2♠.
- NAVIGATION:** Includes a 'Prev' button, a progress slider, a 'Next' button, and a '6 / 99' indicator.
- ACTION LOG:** A section at the bottom for recording actions.

The main area of the dashboard shows a detailed view of a hand. It includes 'ANTE 1' for Round 2 and Round 1. Round 1 shows a 'flush' strategy with cards 9♠, 7♠, 6♠, 5♠, 2♠. Round 2 shows a 'flush' strategy with cards A♠, Q♠, T♠, 7♠, 4♠. The 'HAND HISTORY' section shows the cards played in each hand: A♠, Q♠, T♠, 7♠, 4♠ for Hand 1 and 9♠, 7♠, 6♠, 5♠, 2♠ for Hand 2.

THE LATEST INFRASTRUCTURE

- PPO Model: Learns to choose an overall strategy and make shop decisions.
- Card Heuristics and Monte-Carlo Implementation: Plays the cards.
- Simplified Game Environment: Serves as a smoother, controlled environment, reducing the complexity of the game.
- Parallel training: Allows multiple instances to be trained at the same time, making training even faster.
- Emulator: Allows to train the model on emulator, making it far faster.
- Replay viewer: Allows to see what has occurred at the exact episode.
- Tensorflow graphs: Allow to judge the performance of the model.

RESULTS



- PPO_273 simplified environment, 1 million seeds.
- PPO_274 full environment, 1 millions seeds
- PPO_275 simplified environment, 100 seeds
- PPO_276 full environment, 100 seeds
- PPO_277 simplified environment, 1 seed
- PPO_278 full environment, 1 seed

Models trained on 1 million seeds generalized best, achieving the highest average ante (~4.2).

Models trained on full environment reached higher max ante but lower average ante. This is expected as the simplified environment only has additive jokers and no multiplicative ones.

Simplified version failed to win the game, but the non-simplified version won it by reaching a max ante of 11.

SUMMARY

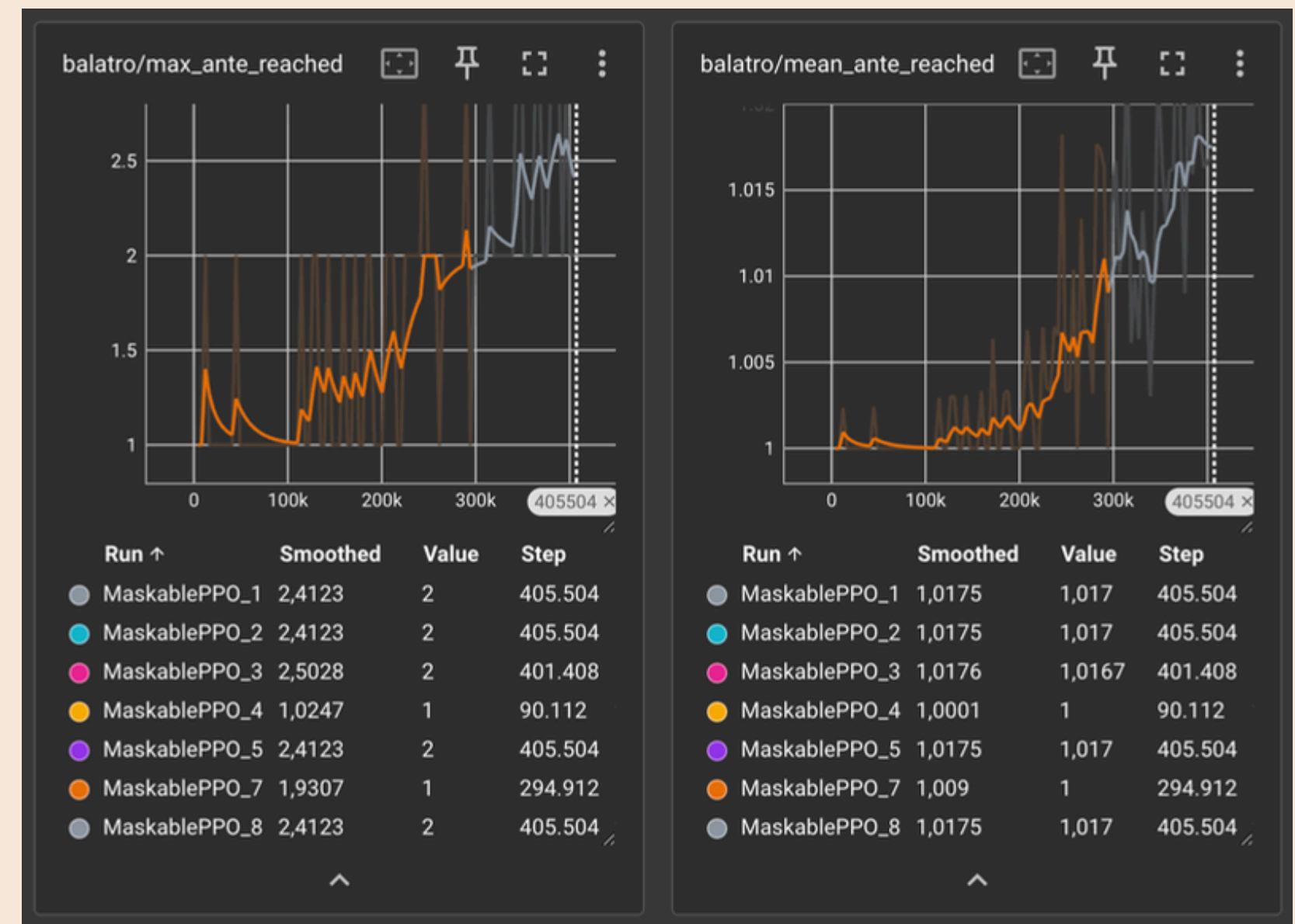
The plateau was an environment ceiling, not an agent one. Allowing model to buy more kinds of jokers improves its performance. By making the environment simpler, we made the game smaller.

Open question: Can the model be further improved?

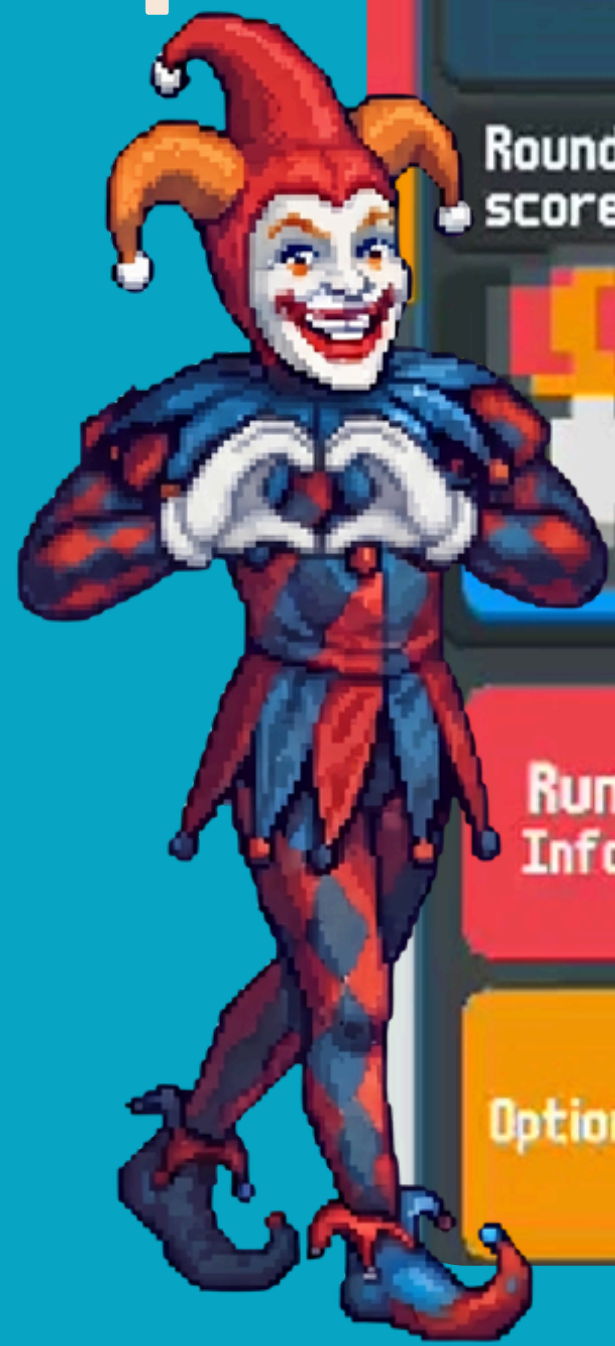
- One could play with different learning rate, batch size, hidden layers or a few other parameters.
- One could allow agent to buy tarrots, planets and other stuff which we believe would further improve the performance of the model.
- Rewards such as selling joker penalty could be played around with.

Answer of the core question: Does our agent really play better than the others?

-Yes, it is reaching both higher max and mean ante number.



THANKS FOR YOUR ATTENTION!



Small Blind

SMALL BLIND

Score at least
❄️ **300**
Reward: \$\$\$

Round score
❄️ 0

0

X

0

Run Info

Options

Hands
4

Discards
4

\$4

Ante
1 / 8

Round
1

0/5

0/2

K♠ Q♠ 10♣ 9♣ 5♣ 5♦ 4♣ 3♥

8/8

Play Hand

Sort Hand

Rank

Suit

Discard